

currencies, Menkoff et al. (2011) and Okunev and White (2000); for real estate, Beracha and Skiba (2011).

7. See Antonacci (2012), Asness et al. (2013), and King, Silver, and Guo (2002).
8. See Fama and French (2008).
9. See <http://papers.ssrn.com/sol3/DisplayAbstractSearch.cfm>.

CHAPTER 3

1. For more in-depth information on the methods of modern finance, see Ilmanen (2011) and Meucci (2009).
2. These assumptions are that portfolio returns are normally distributed or that investors have a quadratic utility function. Quadratic utility implies unrealistic increasing absolute risk aversion, meaning you become more risk averse as your wealth increases. The popular Sharpe ratio relies on these same assumptions.
3. See Ang (2012) and Jacobs, Müller, and Weber (2014).
4. When someone asked Markowitz how he invests his own money, Markowitz said he keeps half in a stock index fund and half in bonds.
5. They were Jack Treynor, William Sharpe, John Lintner, Jon Mossin, and later, Fischer Black.
6. There are now robust methods to correct for autocorrelation and heteroscedasticity. See Newey and West (1987).
7. The lead author, Campbell Harvey, was a former editor of the prestigious *Journal of Finance*. The title of this working paper, "... and the Cross-Section of Expected Returns," is a tongue-in-cheek reference to more than 50 research papers submitted to the journal that included this line in their title, beginning with the seminal one by Fama and French (1992) that started the whole process of factor modeling.
8. A lognormal distribution has fatter tails than a normal distribution and is therefore a better match to the true distribution of stock prices,